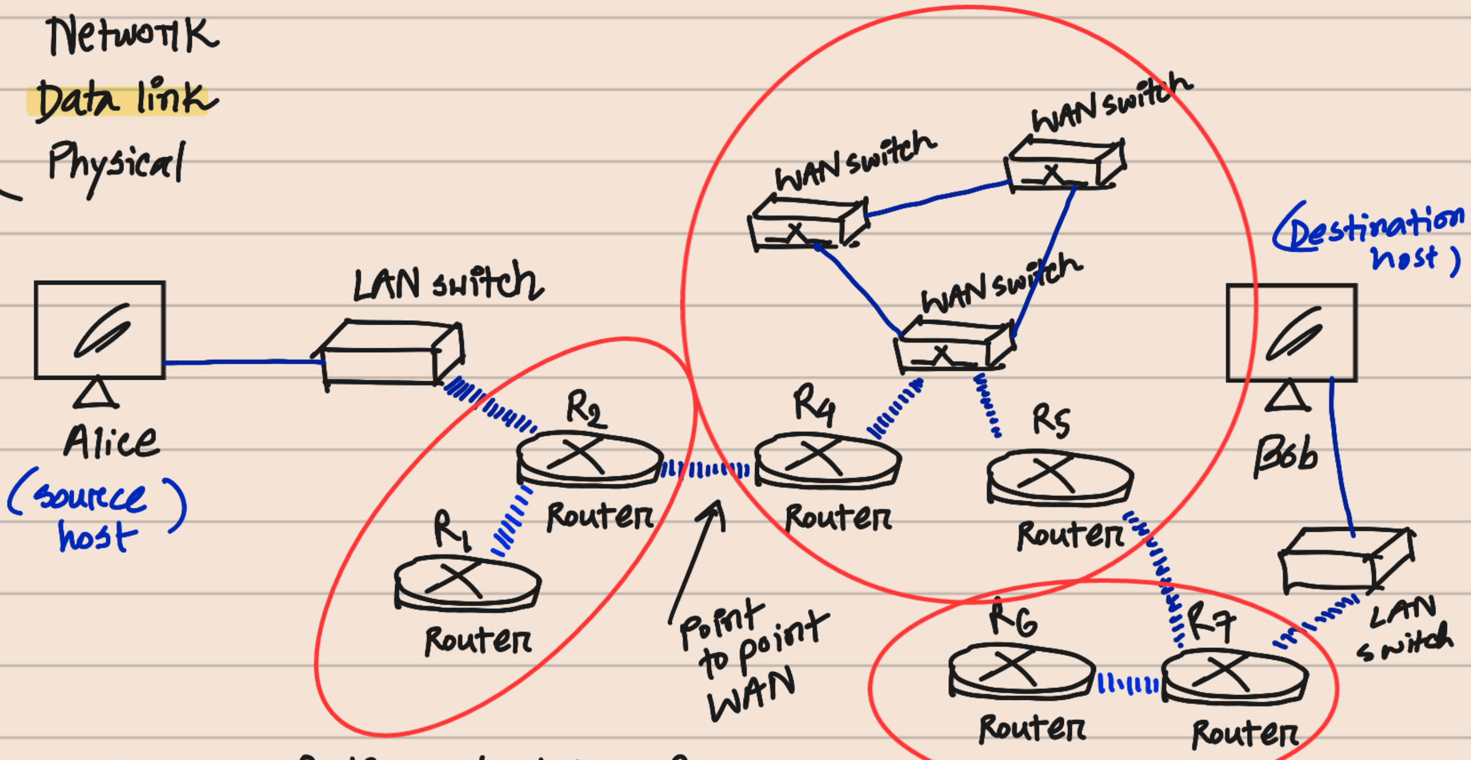
Chapter 9: Introduction to data link layer

- Application
- Transport
- Network
- Data link
- Physical



- Communication at data link layer
  (happens node to node)

[Book diagram]

Nodes – End hosts, routers

Links – Networks in between

Chapter at a glance –

- Data link layer services
  - Framing
  - Flow control
  - Error control
  - Congestion control
- Types of links
  - pt to pt link
  - Broadcast link
- Sublayers
  - Data link Control
  - Media Access Control

- Link layer addressing
  - Reason why IP address is not enough
  - Types of addresses
    (Unicast, Multicast, Broadcast)
- ARP
  - position, operation, packet format

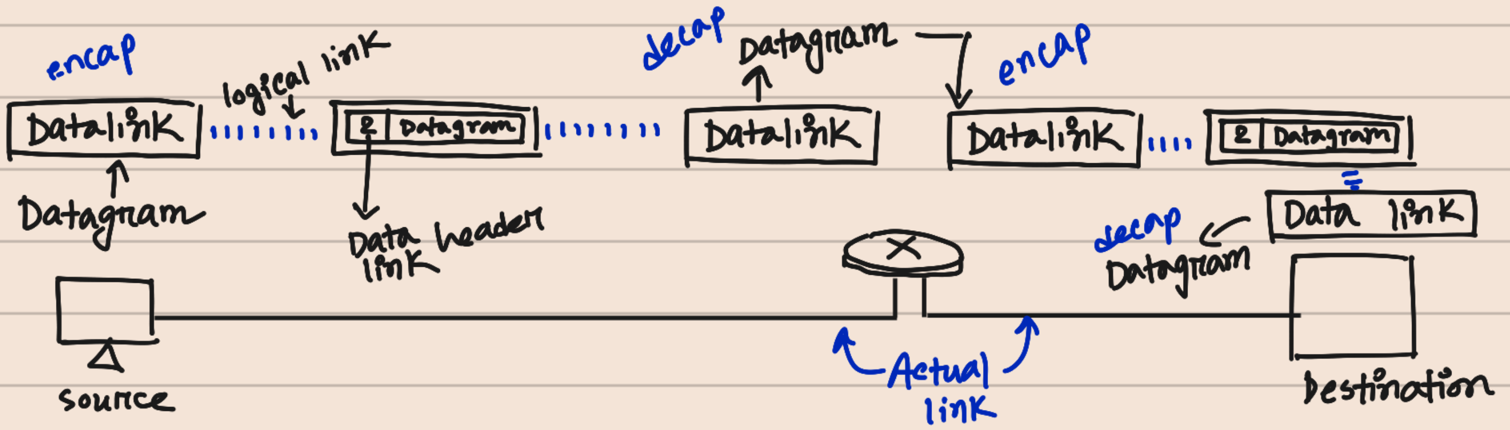
Functions & Services of Data link layer

- Provides service to network layer
  Receives " from physical "

- Delivering a datagram to next node in path
  Sending node — Encapsulates datagram from frame
  Receiving " — Decapsulates
  Intermediate node — Encapsulate + Decapsulate

  Why? Each link may be using a different protocol with different frame format. Even if same protocol, link layer addresses are different.

A datagram is a basic unit of data transfer in packet-switched networks



Services :

1. Framing

   Encapsulation of datagram in a frame before sending to next node
   (packet received from network layer)
   Decapsulation of datagram from frame as well.
   (A packet at data link layer- frame)

2. Flow control

   Producer - Consumer balance

   Producer - Sending data link layer at end of link
   of frame

   Consumer - Receiving " " " " "
   of frame

If rate of produced frames > rate of consumed frames

Frames at receiving end need to be buffered · while waiting

(i) let receiving layer drop frames if buffer full.

or, (ii) receiving layer sending feedback to sender to stop/slow.

3. Error Control

Frame → bits → electromagnetic signal → susceptible to error

- Detect error → corrected / discarded & retransmitted

4. Congestion Control

link may be congested with frames → (result) frame loss

Most data link layer protocols don't use congestion control directly

Types of data link layer

Point to point link	Broadcast link
• Uses whole capacity of medium	• Uses part of capacity
• link dedicated to 2 devices	• Shared communication medium between multiple devices.
• e.g. Landline phone call	• e.g. cellular network
• Simple addressing for data handling	• MAC addressing (link layer addressing)
• Minimal collision chance	• Higher collision chance

Sublayers

Data link control (DLC)
Media Access control (MAC)

Link layer addressing

- IP address — identifiers at network layer
  But for connectionless internetwork (Internet)
  only IP address not enough to send a datagram

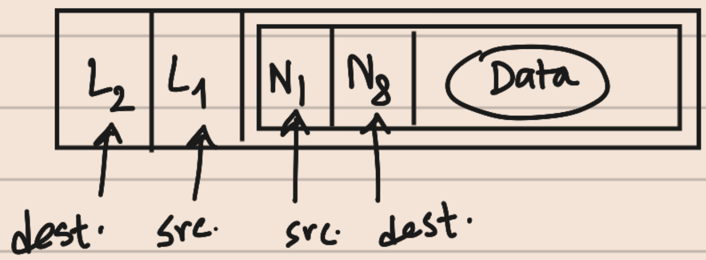
why? Each datagram from src to dest may take a different path
IP address define two ends but cannot define the links to pass through

- IP address should never change.

So, another addressing medium needed :— Link layer addressing
(link address / physical address / MAC address)

Link controlled at data link layer so address need to belong to data link layer.

Frame —



$L \rightarrow$ link layer address
$N \rightarrow$ IP address

- If the IP address of a router does not appear in any datagram sent from a source to a destination, why do we need to assign IP addresses to routers? The answer is that in some protocols a router may act as a sender or receiver of a datagram. For example, in routing protocols we will discuss in Chapters 20 and 21, a router is a sender or a receiver of a message. The communications in these protocols are between routers.
- Why do we need more than one IP address in a router, one for each interface? The answer is that an interface is a connection of a router to a link. We will see that an IP address defines a point in the Internet at which a device is connected. A router with *n* interfaces is connected to the Internet at *n* points. This is the situation of a house at the corner of a street with two gates; each gate has the address related to the corresponding street.
- How are the source and destination IP addresses in a packet determined? The answer is that the host should know its own IP address, which becomes the source IP address in the packet. As we will discuss in Chapter 26, the application layer uses the services of DNS to find the destination address of the packet and passes it to the network layer to be inserted in the packet.
- How are the source and destination link-layer addresses determined for each link? Again, each hop (router or host) should know its own link-layer address, as we discuss later in the chapter. The destination link-layer address is determined by using the Address Resolution Protocol, which we discuss shortly.
- What is the size of link-layer addresses? The answer is that it depends on the protocol used by the link. Although we have only one IP protocol for the whole Internet, we may be using different data-link protocols in different links. This means that we can define the size of the address when we discuss different link-layer protocols.

Types of addresses :

1) Unicast

(NIC)

Each host/interface of router assigned a unicast address

1-1 communication

destination 1.

e.g. in Ethernet 48 bits (6 bytes) → 12 hexadecimal digits

01 : 00 : 5E : ..... (rest decided by other protocol)

2) Multicast

1- many communication (group)

Hexadecimal system

- Address - 1st octet FF : ....

↓ ↓ 8 bit 8 bit

If the last one odd - multicast even - unicast

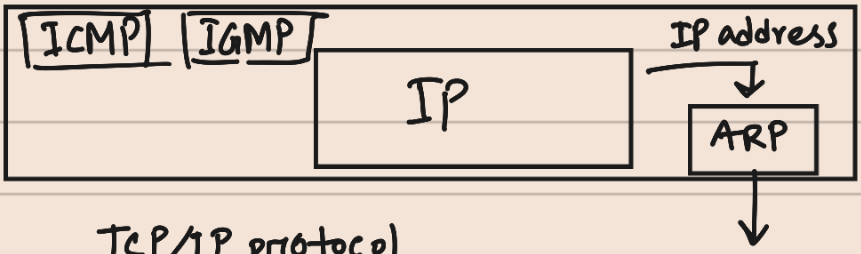
3) Broadcast

1 - all communication

FF : FF : FF : FF : FF : FF

Address Resolution Protocol

- Source host knows – IP address of default router
- Each router except last one in the path gets IP address of next router by using forwarding table.
- Link layer address is needed — ARP helpful.
  It maps an IP address to a logical link address.
- It is an auxiliary protocol (diagnostic/control purpose)



TCP/IP protocol suite

link layer address

Host/Router wants to find link layer address

Sends ARP request packet (sender link layer address & IP address of receiver)

Every host/router on network receives & processes the request

But only intended one recognizes IP and sends back ARP response packet. (recipient link-layer address + IP)

unicast directly to node that sent the request.

ARP packet format

Hardware type		Protocol type
Hardw. length	Protocol length	operation Request : 1, Reply : 2
Source hardware address		
Source protocol address		
Destination hardware address (empty in request)		
Destination protocol address		

Hardware – LAN/WAN protocol
Protocol – Network layer "

Packet Format

Figure 9.8 shows the format of an ARP packet. The names of the fields are self-explanatory. The *hardware type* field defines the type of the link-layer protocol; Ethernet is given the type 1. The *protocol type* field defines the network-layer protocol: IPv4 protocol is $(0800)_{16}$. The source hardware and source protocol addresses are variable-length fields defining the link-layer and network-layer addresses of the sender. The destination hardware address and destination protocol address fields define the receiver link-layer and network-layer addresses. An ARP packet is encapsulated directly into a data-link frame. The frame needs to have a field to show that the payload belongs to the ARP and not to the network-layer datagram.